Light

GCSE Computer Science — Boolean Logic (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What output does an AND gate produce when both inputs are 1?

- A. 0
- B. 1
- C. 2
- D. Undefined

Q2 (1 mark)

What output does an OR gate produce when at least one input is 1?

- A. 0

- ○ **B.** The opposite of the other input
- ○ **C.** The same as the other input
- ○ **D.** 1

Q3 (1 mark)

What does a NOT gate do to its input?

- ○ **A.** Doubles the input value
- ○ **B.** Adds 1 to the input
- ○ **C.** Keeps the input unchanged
- ○ **D.** Inverts the input (changes 0 to 1, or 1 to 0)

Q4 (1 mark)

In an AND gate truth table, how many input combinations produce an output of 1?

- ○ **A.** 1 combination
- ○ **B.** 2 combinations
- ○ **C.** 3 combinations
- ○ **D.** 4 combinations

Q5 (1 mark)

In an OR gate truth table, how many input combinations produce an output of 1?

- ○ **A.** 1 combination
- ○ **B.** 2 combinations
- ○ **C.** 3 combinations
- ○ **D.** 4 combinations

Q6 (1 mark)

Which logic gate symbol has a curved back and a pointed front (like a bullet shape)?

- ○ **A.** NOT gate
- ○ **B.** AND gate
- ○ **C.** OR gate
- ○ **D.** There is no such symbol

Q7 (1 mark)

What small symbol appears at the output of a NOT gate to indicate inversion?

- ○ **A.** A small circle (bubble)
- ○ **B.** A small square
- ○ **C.** A small triangle

- ○ **D.** A small arrow

Q8 (1 mark)

In Boolean logic truth tables, what values can inputs and outputs take?

- ○ **A.** Any number from 0 to 9
- ○ **B.** Only 0 and 1 (or equivalently True/False)
- ○ **C.** Only positive numbers
- ○ **D.** Any letter of the alphabet

Q9 (1 mark)

A security system requires BOTH a valid keycard AND a correct PIN to unlock. Which logic gate represents this requirement?

- ○ **A.** OR gate
- ○ **B.** NOT gate
- ○ **C.** AND gate
- ○ **D.** None of the above

Q10 (1 mark)

A car alarm activates if the door sensor OR the motion sensor detects intrusion. Which logic gate represents this behaviour?

- ○ **A.** AND gate

- B. No gate is needed
- C. NOT gate
- D. OR gate

Q11 (1 mark)

A night light turns ON when the room is dark (light sensor = 0). Which gate converts 'dark' (0) to 'light ON' (1)?

- A. AND gate
- B. OR gate
- C. NOT gate
- D. No gate is needed

Q12 (1 mark)

How many rows does a truth table have for a logic circuit with 3 inputs?

- A. 3 rows
- B. 6 rows
- C. 8 rows
- D. 9 rows

Q13 (2 marks)

What is the output of NOT (1 AND 0)?

- A. 0
- B. 1
- C. Cannot be determined
- D. 2

Q14 (2 marks)

What is the output of NOT (0 OR 1)?

- A. 0
- B. 1
- C. Depends on other factors
- D. Neither 0 nor 1

Q15 (2 marks)

Evaluate: (1 AND 1) OR (0 AND 1)

- A. 0
- B. 11
- C. 10
- D. 1

Q16 (2 marks)

Evaluate: (NOT 1) AND (NOT 0)

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** Undefined
- ○ **D.** -1

Q17 (2 marks)

In a truth table for A AND B, which row shows: A=0, B=1, Output=?
What is the missing output value?

- ○ **A.** 1
- ○ **B.** 0
- ○ **C.** A
- ○ **D.** B

Q18 (2 marks)

In a truth table for A OR B, which row shows: A=0, B=0, Output=?
What is the missing output value?

- ○ **A.** A OR B
- ○ **B.** 1
- ○ **C.** 0

- **D.** Undefined

Q19 (2 marks)

A logic circuit has inputs A and B. The output is described as 'A AND (NOT B)'. If A=1 and B=1, what is the output?

- **A.** 0
- **B.** 1
- **C.** Cannot be calculated
- **D.** Depends on the gate order

Q20 (2 marks)

Which Boolean expression describes a circuit where the output is 1 only when input A is 1 AND input B is 0?

- **A.** A OR B
- **B.** A AND B
- **C.** A AND (NOT B)
- **D.** NOT A AND B

Q21 (2 marks)

Evaluate: (A OR B) AND C, where A=1, B=0, C=1

- **A.** 0

- ○ **B.** Cannot be determined
- ○ **C.** 2
- ○ **D.** 1

Q22 (2 marks)

NOT (A AND B) produces the same output as which expression when tested with all input combinations?

- ○ **A.** NOT A AND NOT B
- ○ **B.** NOT A OR NOT B
- ○ **C.** A OR B
- ○ **D.** A AND B

Q23 (2 marks)

A heating system turns on (output=1) when the temperature is low (sensor=0) AND the system is enabled (switch=1). Which expression represents this?

- ○ **A.** sensor AND switch
- ○ **B.** (NOT sensor) AND switch
- ○ **C.** sensor OR switch
- ○ **D.** (NOT sensor) OR switch

Q24 (3 marks)

A logic circuit has two inputs P and Q. The complete truth table is:

P	Q	Output
0	0	1
0	1	0
1	0	0
1	1	0

Which Boolean expression produces this truth table?

- ○ **A.** P AND Q
- ○ **B.** P OR Q
- ○ **C.** NOT P AND NOT Q
- ○ **D.** NOT (P OR Q)

Q25 (3 marks)

A logic circuit has three inputs A, B, and C. The circuit is: (A AND B) OR (NOT C).
For which input combination does the output equal 0?

- ○ **A.** A=1, B=1, C=0
- ○ **B.** A=0, B=0, C=0
- ○ **C.** A=0, B=0, C=1
- ○ **D.** A=1, B=0, C=0

Q26 (3 marks)

A smart home system controls a fan. The fan turns ON when:

- The temperature is high (T=1) AND the room is occupied (O=1), OR
- Manual override is activated (M=1)

Which expression correctly models this system?

- ○ **A.** (T OR O) AND M
- ○ **B.** (T AND O) OR M
- ○ **C.** T AND O AND M
- ○ **D.** T OR O OR M

Q27 (3 marks)

Complete the truth table for: (A OR B) AND (NOT B)

A	B	Output
0	0	?
0	1	?
1	0	?
1	1	?

Which row pattern shows the correct outputs (in order from top to bottom)?

- ○ **A.** 0, 0, 0, 0
- ○ **B.** 1, 1, 1, 1
- ○ **C.** 1, 0, 1, 0
- ○ **D.** 0, 0, 1, 0

Q28 (3 marks)

A student creates a logic circuit intended to output 1 only when exactly one of inputs A or B is 1 (but not both). They use: A OR B
What is wrong with this circuit, and what happens when A=1 and B=1?

- ○ **A.** Nothing is wrong; the circuit works correctly and outputs 0
- ○ **B.** The circuit incorrectly outputs 1 when both inputs are 1, because OR doesn't distinguish between 'at least one' and 'exactly one'
- ○ **C.** The circuit outputs 0 when both inputs are 0, which is incorrect
- ○ **D.** The circuit needs more inputs to work properly

Q29 (3 marks)

A bank vault door opens when: (Manager keycard AND Employee keycard) OR (Emergency override AND NOT Regular hours).
During regular hours (R=1), the manager has their card (M=1), the employee has their card (E=1), and emergency override is active (O=1). Does the door open?

- ○ **A.** Yes, because both keycards are present
- ○ **B.** No, because emergency override cancels the keycards
- ○ **C.** Yes, because of the emergency override
- ○ **D.** No, because regular hours blocks both conditions

Q30 (3 marks)

A logic circuit uses the expression: (A AND B) OR (A AND NOT B). Analyse this expression. What simpler equivalent expression produces the same outputs?

- **A.** A OR B
- **B.** A AND B
- **C.** A
- **D.** B

Total: **55** marks • Time: **30 min**

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